



A comparative overview of frequency domain methods for nonlinear systems☆☆



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ABSTRACT

The widespread acceptance of frequency domain techniques for linear and time invariant systems has been an impetus for the extension of these methodologies toward nonlinear systems. However, differences and equivalences between alternative methods have been less addressed. This paper provides a comparative overview of four classes of frequency domain methods for nonlinear systems: Volterra based models, nonlinear frequency response functions / Bode plots, describing functions and linear approximations in the presence of nonlinearities. Each method is introduced using consistent nomenclature and terminology, which allows for comparison in terms of system and signal classes for which the methods are valid as well as the type of (nonlinear) effects captured by each model. Summarizing, the paper aims to connect, and make different frequency domain methods for nonlinear systems accessible, by providing a comparative overview of such methodologies, accompanied by illustrative (experimental) examples.

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1. Introduction

For linear and time invariant (LTI) systems, frequency domain techniques resulted in a widespread acceptance in the engineering community for analysis, modeling and controller design [9]. Correspondingly, the Frequency Response Function (FRF) and representations such as the Bode, Nyquist and Nichols plot have become standard engineering tools and have proven to be indispensable for the modeling and design of dynamic systems in industry. Increased performance requirements of these systems result in a different attitude toward nonlinear system behavior. Previously a cer-

tain degree of nonlinear behavior could be ignored because it did not impair system performance. In many present day dynamic systems however nonlinear behavior limits performance or is deliberately introduced to improve performance. See for example [45,89]. In both cases nonlinear effects can no longer be neglected. Hence, the widespread acceptance of frequency domain techniques for LTI systems has been an impetus for the extension of these methodologies toward nonlinear systems.

Although different approaches have been independently developed to analyze and model nonlinear systems in the frequency domain, differences and equivalences between alternative methods have been minorly addressed. In this paper an overview and comparison of the following four well established approaches is presented:

- Volterra series based approaches (Section 3)
- Frequency response function and Bode plot for nonlinear systems (Section 4)
- Describing functions (Section 5)
- Linear approximations in the presence of nonlinearities (Section 6)

Table 1 provides a comparative overview of the main characteristics of these modeling approaches and constitutes one of the

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Table 1

Overview of frequency domain modeling approaches for nonlinear systems. Legend: dark grey: effect captured / information available, light grey: effect partially captured / requires additional processing, white: effect not captured / information not available. For a description of the different frequency domain effects, see [98].

		captured nonlinear effect				information content			
		gain compression / expansion	desensitization	intermodulation	harmonics	at frequencies present in input		at frequencies not present in input	
		dependence of the gain on the input excitation level	dependence of the response on the input at another frequency	input frequencies combine to produce new frequencies in the output	generation of frequency components in the output at multiples of the input frequency	gain	phase	gain	phase
conventional FRF									
generalized FRF									
nonlinear FRF									
describing funct.	sinusoidal								
	generalized								
	HOSIDF								
best linear appr.									

main results of this paper. This comparison is further motivated in the sequel.

First, a brief introduction to the effects of nonlinearities in the frequency domain is presented in Section 2. Next, Sections 3 – 6 introduce each model class. This includes a definition of the system and signal class for which the model types are defined. The notation and terminology used in the different sections are consistent, which allows for comparison between different approaches.

A brief overview of relevant literature is provided for each method (see Table 4, Appendix B, for a summary). Moreover, each section is concluded by either an analytical example or an experimental case study, illustrating the introduced approach. Finally, in Section 7 a comparison of the properties of the different model types is provided as well as an assessment of their applicability in practice. Finally, note that the required nomenclature and preliminaries are presented in Appendix A to increase readability of the paper.

2. Nonlinearities in the frequency domain

A fundamental property of LTI systems is that it cannot shift energy from one frequency to the other. Hence, the response to a sinusoidal input with a particular frequency is again sinusoidal with the same frequency as the input signal. The phase shift and gain relating the input and output are characterized by the FRF at that particular frequency and as superposition holds for LTI systems, the response to more general input signals is fully captured by the FRF as well. However, for nonlinear systems the superposition principle does not hold and the response to even a simple sinusoid can be a complex, multi harmonic signal. However, the increased richness in the output allows for effective identification and reduction/utilization of nonlinear effects. See for example [23,70,72,74,85]. Hence, classical frequency domain approaches for LTI systems cannot be straightforwardly applied when nonlinearities are present and additional analysis is required to investigate if, when and how similar methodologies can be used when analyzing nonlinear systems.

In the frequency domain, nonlinear effects manifest themselves in different ways. The gain of a nonlinear system may, for example, depend on the amplitude of the input and the output may contain

harmonics of the input frequency. Moreover, when a nonlinear system is subject to a multisine input, input frequencies may combine, producing new frequencies which are not present in the input signal and spectral components at a given frequency in the output may depend on other frequencies in the input. These effects are summarized in Table 1 and they are used in the following to discuss different frequency domain models for nonlinear systems. To further illustrate the effects listed in Table 1, consider the following example.

Example 1 (effects of nonlinearities in the frequency domain). Consider the following harmonic signal with nonzero spectral contributions at 1 [Hz] and 5 [Hz].

$$u(t) = a \cos(1 \cdot 2\pi t) + b \cos(5 \cdot 2\pi t)$$

Now consider a static nonlinear mapping $y(t) = 4u^3(t)$, which yields:

$$\begin{aligned}
 y(t) = & \underbrace{(3a^3 + 6ab^2) \cos(1 \cdot 2\pi t) + (a^3 + 3a^2b) \cos(3 \cdot 2\pi t)}_{\text{gain comp./exp. \& desensitization harmonics \& intermodulation}} \\
 & + \underbrace{(6a^2b + 3b^3) \cos(5 \cdot 2\pi t) + 3a^2b \cos(7 \cdot 2\pi t)}_{\text{gain comp./exp. \& desensitization intermodulation}} \\
 & + \underbrace{3ab^2 \cos(9 \cdot 2\pi t) + 3ab^2 \cos(11 \cdot 2\pi t)}_{\text{intermodulation intermodulation}} \\
 & + \underbrace{b^3 \cos(15 \cdot 2\pi t)}_{\text{harmonics}}
 \end{aligned} \tag{1}$$

The expression in (1) clearly shows the different effects indicated in Table 1. First of all, multiples of the input frequencies 1 [Hz] and 5 [Hz] appear at 3 [Hz] and 15 [Hz] (harmonics). Moreover, the 1 [Hz] and 5 [Hz] components in (1) depend nonlinearly on the input amplitude (gain compression / expansion) and depend on the input at other frequencies as well (desensitization). Finally, the nonlinear mapping combines input frequencies and their sum and difference frequencies to new frequencies at 3, 7, 9 and 11 [Hz] (intermodulation).

These effects are visualized in Fig. 1 where the nonlinear deformation of the signal $u(t)$ is depicted for $a = 2$ and $b = 3$. Fig. 1a and b show the deformation of the signal in the time domain. The

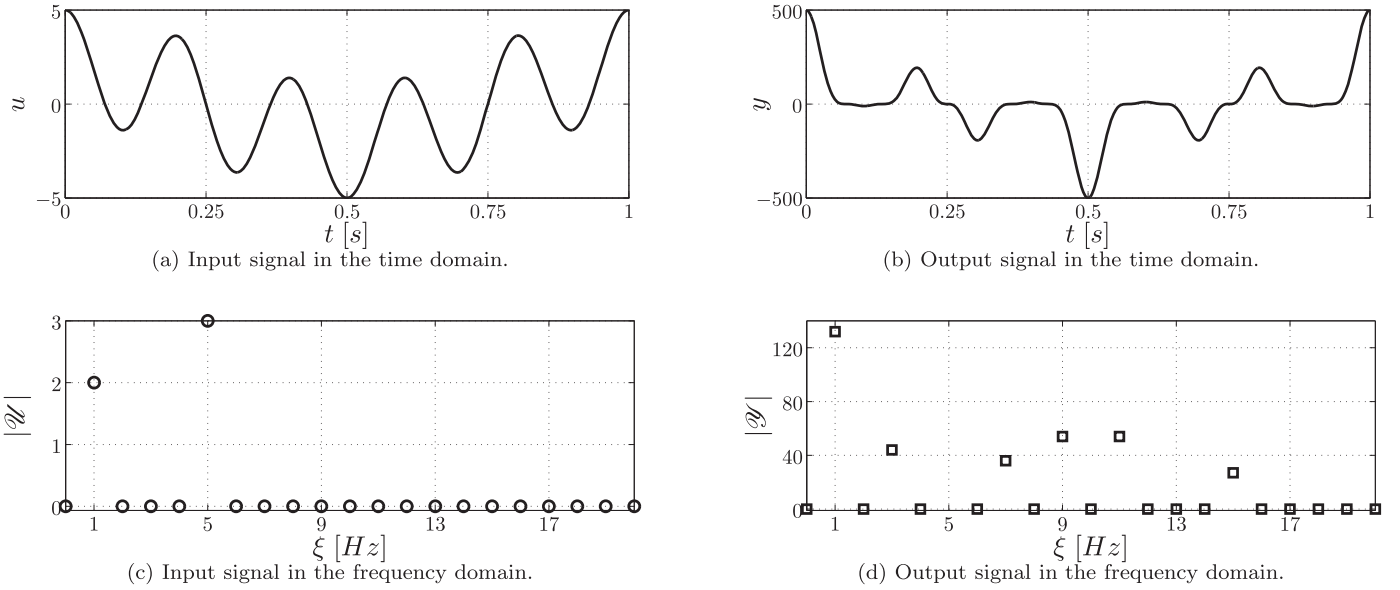


Fig. 1. Time and frequency domain representation of a nonlinearly deformed harmonic signal.

corresponding frequency domain representations are depicted in Fig. 1c and d. For a description of the different frequency domain effects, see also [98].

Before starting the formal introduction, application and comparison of frequency domain methods that aim to capture these nonlinear effects one may read Appendix A, which introduces the necessary nomenclature. Next, the first of the four model classes, i.e. the Volterra series based approach, is addressed.

3. Volterra series based approaches

Modeling of nonlinear phenomena is often achieved by identifying a local approximation of the behavior by an approximative series. In case of continuous functions, a polynomial or Taylor approximation is often used and the existence and convergence of such series is analyzed in the Stone–Weierstrass theorem [87,88,94]. An extension of the series approximation for nonlinear dynamical systems is based on the work of Vito Volterra [92,93] which was further developed by Norbert Wiener during World War II [95]. This series approximation is referred to as the Volterra series [76,78]. In this section, the frequency domain representation of the Volterra series developed in the 1950s [18] is discussed, which is referred to as the generalized frequency response function.

The Volterra series approximation generalizes the polynomial approximation of 'static functions' over a given interval, to the approximation of dynamical systems around a given working point [27]. This series expansion captures the input-output dynamics of a (nonlinear) dynamical system in a set of Volterra kernels that are a generalization of the impulse response of LTI systems. Provided that a Volterra series exists and can be identified [10,21,39,77], the response up to order $\wp$, i.e. $y_{\wp}(t) \in \mathbb{R}$ of such system to an input $u(t) \in \mathbb{R}$ is given by:

$$y_p(t) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} h_p(\tau_1, \tau_2, \dots, \tau_p) \prod_{m=1}^p u(t - \tau_m) d\tau_m$$

$$y_{\wp}(t) = \sum_{p=1}^{\wp} y_p(t) \quad (2)$$

where $h_p(\tau_1, \tau_2, \dots, \tau_p) : \mathbb{R}^p \mapsto \mathbb{R}$ is the p^{th} order Volterra kernel and $y_p(t)$ is referred to as the corresponding p^{th} order output. The Volterra series model (2) is defined for the class of Volterra systems, which is defined as follows.

Definition 1 ($\nabla \mathbb{S}$: Volterra system). A dynamical system is called a Volterra system if there exists a Volterra series representation (2) that converges uniformly around a given working point, with probability 1 to $y(t)$ for all bounded inputs $u \in \mathbb{G}$, i.e.

$$\lim_{\wp \rightarrow \infty} \int_0^{\tau} E\{|y(t) - y_{\wp}(t)|\} = 0 \quad \forall \tau \in \mathbb{R}_{\geq 0}$$

where the expected value $E\{\cdot\}$ is the ensemble average over the considered class of random inputs.

(see: [78,82])

The frequency domain representation corresponding to (2) is obtained by applying the multiple Fourier transform to the p^{th} order Volterra kernel. This yields the definition of the p^{th} order Generalized Frequency Response Function (GFRF).

Definition 2 ($\mathfrak{F}_p(\varpi_p)$: Generalized frequency response function (GFRF)). Consider a Volterra system according to Definition 1. Then, its p^{th} order GFRF is denoted $\mathfrak{F}_p(\varpi_p) : \mathbb{R}^p \mapsto \mathbb{C}$, with $\varpi_p = (\xi_1, \xi_2, \dots, \xi_p) \in \mathbb{R}^p$. The GFRF is defined as the p -dimensional Fourier transform of the p^{th} order Volterra kernel in (2), i.e.

$$\mathfrak{F}_p(\varpi_p) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} h_p(\tau_1, \dots, \tau_p) \prod_{m=1}^p e^{-2\pi i \xi_m \tau_m} d\tau_m \quad (3)$$

(see: [6,16,18,78])

The concept of the GFRF was first introduced in [18] as a frequency domain equivalent of the Volterra kernel, and represents a generalization of the FRF for LTI systems. Although identification of the FRF for LTI systems is well understood, identification of the GFRF is nontrivial. Nevertheless, several approaches exist to obtain the GFRF for different classes of nonlinear systems. First of all, for some nonlinear models the GFRF can be computed analytically. Examples of such approaches are: rational nonlinear models [100], nonlinear, autoregressive moving average models with exogenous inputs (NARMAX) [58] and nonlinear integro-differential equations [5]. Moreover, the GFRF of a known NARMAX model may also be obtained by application of the probing method [2,6,8] and

more complex dynamical systems may be modeled by a series of piecewise valid GFRFs [42]. Finally, the GFRF of quadratic and cubic nonlinear systems can be obtained in a numerically efficient manner by application of a symbolic expression of the output of such system as discussed in [41].

In general, a parametric representation is often used to analyze the GFRF. For example, in [33], a parameter extraction operator is discussed that allows to study the influence of the parameters of a nonlinear model on the GFRFs. This operator is applied to analyze the relationship between different order GFRFs [28] and to distinguish between input and output nonlinearities in a class nonlinear models [31]. Moreover, it is used to characterize and efficiently compute the generalized output frequency response function as discussed in [29,30,57]. Where the GFRF describes the relation between the input and output of Volterra systems, the generalized output frequency response function relates the parameters of a nonlinear model to the output spectrum of the nonlinear system [36–38].

For Volterra systems, the GFRF allows for spectral analysis of the nonlinear dynamics. In [35], it is shown that it is possible to predict the output frequency range for an arbitrary input signal using the GFRF. Moreover, a spectral analysis of block structured dynamical systems in terms of the GFRF is presented in [26] and the GFRF can be used to attain a user defined frequency domain performance for a nonlinear system as shown in [32,34]. Finally, application of the GFRF allows to compute bounds on the output of Volterra systems [3,4] and assess convergence of the corresponding Volterra series [21,40].

Although the GFRF allows for detailed spectral analysis of the behavior of Volterra systems their interpretation is nontrivial. In [98] an interpretation of the nonlinear effects in Table 1 in terms of the GFRF is presented. Moreover, techniques to visualize the GFRFs up to order three are introduced in [7,99]. Finally, to illustrate the derivation of the GFRF the following example, taken from [5], presents the computation of the GFRF for the Duffing oscillator (continued in [24,43]).

Example 2 (GFRF of the forced Duffing oscillator [5]). The GFRFs can be analytically computed for nonlinear systems described by nonlinear integro-differential equations. A well known example of such nonlinear system is the Duffing oscillator which dynamics are given by:

$$\ddot{y}(t) + 2\zeta\omega_n\dot{y} + \omega_n^2 y(t) + \omega_n^2 \varepsilon y^3(t) = u(t) \quad (4)$$

with $\zeta, \omega_n, \varepsilon \in \mathbb{R}$ and the input and output are denoted by $u(t)$ and $y(t)$ respectively. As shown in [5] the first three GFRFs of the system in (4) equal:

$$\begin{aligned} \mathfrak{T}_1(\xi_1) &= \frac{1}{\omega_n^2 + 4\pi\zeta\omega_n i\xi_1 - 4\pi^2\xi_1^2} \\ \mathfrak{T}_2(\xi_1, \xi_2) &= 0 \\ \mathfrak{T}_3(\xi_1, \xi_2, \xi_3) &= -\varepsilon\omega_n^2 \mathfrak{T}_1\left(\sum_{\ell=1}^3 \xi_\ell\right) \prod_{m=1}^3 \mathfrak{T}_1(\xi_m) \end{aligned}$$

where $\mathfrak{T}_1(\xi_1)$ relates to the linear part of the dynamics in (4) and the cubic nonlinearity appears in $\mathfrak{T}_3(\xi_1, \xi_2, \xi_3)$.

4. Nonlinear frequency response function and Bode plot

In this section a class of frequency response functions for nonlinear systems is introduced that is based on the concept of convergent systems. The idea of convergent systems originates from

Russian literature [12] and relates to the existence of stable and unique limit solutions of nonlinear systems. The notion of convergent systems is introduced in [50,54] as follows:

Definition 3 ($\overline{\mathcal{CS}}$: Uniformly convergent systems). A time invariant system (11) is said to be uniformly convergent for a class of input signals $\mathfrak{W}$ if for every $w \in \mathfrak{W}$:

1. All solutions $x_w(t)$ are well-defined for all $t \in [t_0, \infty)$ and all initial conditions $x_0 \in \mathbb{R}^n$;
2. There exists a unique solution $\tilde{x}_w(t)$ defined and bounded for all $t \in (-\infty, +\infty)$;
3. The solution $\tilde{x}_w(t)$ is uniformly globally asymptotically stable.

To emphasize the dependence on the input $w(t)$, the limit solution is denoted by $\tilde{x}_w(t)$.

(see: [50,54])

In [51,53] convergence is discussed for interconnected and piecewise affine systems, which allows to analyze tracking and synchronization in this type of nonlinear system [97]. For uniformly convergent nonlinear systems, it is possible to relate a sinusoidal input signal to the corresponding steady-state solution and output of the system. These mappings are referred to as the nonlinear state and output frequency response function (NSFRF and NOFRF) and are collectively referred to as the nonlinear frequency response function (NFRF). The NFRF was first introduced in [55] and is defined as follows:

Definition 4 ($\mathfrak{N}_{S/O}(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega)$: Nonlinear state and output FRF (NFRF)). Consider the system (11) subject to a sinusoidal input¹ $w(t) = \gamma \sin(\omega t)$ with frequency $\xi = \omega/2\pi$ and amplitude γ and assume that:

1. $f(x, w)$ is locally Lipschitz with respect to x ;
2. The system is uniformly convergent for the class of bounded continuous input signals;
3. The system is input to state stable.

Then, there exist a continuous function $\mathfrak{N}_S(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega) : \mathbb{R}^3 \mapsto \mathbb{R}^n$, called the Nonlinear State Frequency Response Function (NSFRF), such that for any sinusoidal input the steady state solution of (11) is given by:

$$\tilde{x}_w(t) = \mathfrak{N}_S(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega) \quad (5)$$

Furthermore, the Nonlinear Output Frequency Response Function (NOFRF) is denoted by $\mathfrak{N}_O(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega) : \mathbb{R}^3 \mapsto \mathbb{R}^m$ and relates a sinusoidal input to the corresponding steady state output. Using the definition of the NSFRF in (5), the NOFRF is defined as:

$$\mathfrak{N}_O(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega) = g(\mathfrak{N}_S(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega))$$

(see: [55])

Numerically efficient methods are required to compute the periodic solutions required to derive the NFRF. In [1,52], examples of such methods are presented. When using the NFRF, a method for visualization of the dynamics is required for analysis and controller design. For LTI systems, the Bode plot visualizes the FRF and is widely applied in both analysis and controller design. Using the NOFRF, the concept of the Bode plot is extended to nonlinear systems in [55]. Therefore, consider the following definition of the nonlinear Bode plot.

Definition 5 ($\mathfrak{B}(\omega, \gamma)$: Nonlinear Bode plot). Consider the system (11), with scalar output $y(t) \in \mathbb{R}$ and subject to a sinusoidal input $w \in \mathbb{S}$ with frequency $\xi = \omega/2\pi$ and assume conditions 1–3 in

¹ In the remainder of this section the angular frequency $\omega = 2\pi\xi$ is used for ease of notation.

Definition 4 are satisfied. Then, the nonlinear Bode plot $\mathfrak{G}(\omega, \gamma) : \mathbb{R}^2 \mapsto \mathbb{R}$ is defined as:

$$\mathfrak{G}(\omega, \gamma) = \frac{1}{\gamma} \left(\sup_{t \in [-\frac{\pi}{\omega}, \frac{\pi}{\omega}]} |\mathfrak{N}_0(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega)| \right)$$

where $\mathfrak{N}_0(\cdot)$ is the NOFRF as in **Definition 4**.

(see: [55])

Remark. The nonlinear Bode plot represents a conservative estimate of the system gain in the sense that the ratio between the input amplitude and the spectral component of the output at the input frequency may be smaller than the value shown in the nonlinear Bode plot.

The definition of the nonlinear Bode plot in **Definition 5**, although based on the norm of a time domain signal, relates directly to the gain of the FRF as visualized in the classical Bode plot for LTI systems. However, where the classical Bode plot provides information about the phase shift as well, such phase information cannot be defined for nonlinear systems. Moreover, where system norms can be computed from the FRF and Bode plot for LTI systems, this is generally not possible for nonlinear systems based on the NFRF and nonlinear Bode plot. The concept of the nonlinear Bode plot is used to extend the notion of the sensitivity function to nonlinear systems in [56]. These generalized sensitivity functions have been applied for the design of variable gain control for optical storage drives and wafer scanners [25,96] and feed forward design for wafer scanners [22]. Finally, the following example is taken from [55] and concludes this section by illustrating the computation of the NFRF and nonlinear Bode plot for a given nonlinear system.

Example 3 (nonlinear state and output FRF [55]). Consider the following dynamical system, subject to a sinusoidal input $u(t) = \gamma \sin(\omega t)$ with frequency $\xi = \omega/2\pi$ and amplitude γ .

$$\begin{aligned} \dot{x}_1 &= -x_1 + x_2^2 & y &= x_1 \\ \dot{x}_2 &= -x_2 + u \end{aligned} \quad (6)$$

As shown in [55], condition 1–3 in **Definition 4** are satisfied for this system. Hence, the NSFRF exists and is unique. Next, using the results from [55] and denoting $v(t) = [\sin(\omega t) \cos(\omega t)]^T$, the NSFRF, which describes the steady state solution $\tilde{x}_u(t) = [\tilde{x}_{1,u}(t) \tilde{x}_{2,u}(t)]^T$ of (6), equals:

$$\mathfrak{N}_S(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega) = \gamma \begin{bmatrix} \gamma v^T(t) \Psi_1(\omega) v(t) \\ \Psi_2(\omega) v(t) \end{bmatrix}$$

$$\Psi_1(\omega) = (1 + \omega^2)^{-2} (1 + 4\omega^2)^{-1} \begin{bmatrix} 2\omega^4 + 1 & \omega^3 - 2\omega \\ \omega^3 - 2\omega & 2\omega^4 + 5\omega^2 \end{bmatrix}$$

$$\Psi_2(\omega) = (1 + \omega^2)^{-1} \begin{bmatrix} 1 & -\omega \end{bmatrix}$$

Using these results, and as $y(t) = g(x(t)) = x_1(t)$, the corresponding NOFRF, which describes the output corresponding to the steady state solution $\tilde{y}_u(t) = \tilde{x}_{1,u}(t)$ of (6), equals:

$$\mathfrak{N}_0(\gamma \sin(\omega t), \gamma \cos(\omega t), \omega) = \gamma^2 v^T(t) \Psi_1(\omega) v(t)$$

Finally, the nonlinear Bode plot is obtained by (numerically) computing:

$$\mathfrak{G}(\omega, \gamma) = \frac{1}{\gamma} \left(\sup_{t \in [-\frac{\pi}{\omega}, \frac{\pi}{\omega}]} |\gamma^2 v^T(t) \Psi_1(\omega) v(t)| \right) \quad (7)$$

and the result is depicted in Fig. 2.

Summarizing, the NSFRF relates any sinusoidal input to the corresponding steady state solution and the NOFRF yields the corresponding steady state output. Finally, the nonlinear Bode plot yields the maximum gain of the nonlinear system for a sinusoidal input with a given amplitude and excitation frequency.

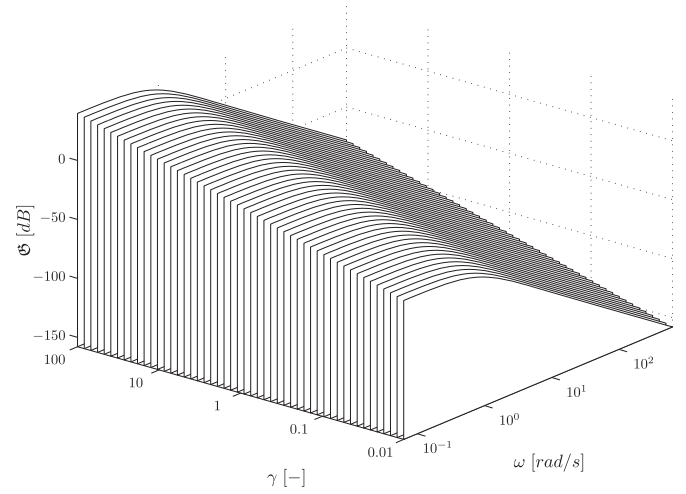


Fig. 2. Nonlinear Bode plot (7) of system (6).

5. Describing functions

For LTI systems the FRF fully captures the systems dynamics as the response to a single sinusoid can be extended to arbitrary inputs using the principle of superposition. For nonlinear systems superposition does not hold and the FRF fails to capture the complete dynamics. Hence, an alternative approach is required to extend the concept of the FRF to nonlinear systems. Describing Functions (DF) aim to do so by describing the response to a given type of input signal based on some frequency domain mapping from input to output. Although different types of describing functions exist (Fig. 3), their definition is generally similar to that of the FRF for LTI systems. The main difference between different describing functions originates from the signal and system class for which they are defined and the type of nonlinear effects captured. In this section a selection of describing functions is introduced, compared and an experimental example is provided illustrating the identification of one of the describing functions.

In Fig. 3, three classes of describing functions are depicted: the sinusoidal describing function, the generalized describing function and the higher order sinusoidal describing function. The different branches represent different subclasses of DF such that for any two consecutive DF, the lower one is a special case of the upper one. The classical (two) sinusoidal input describing function (SIDF) models the response to (two) sinusoidal inputs at the corresponding excitation frequencies. The Generalized Describing Function (GDF) on the other hand, is based on the Volterra series approximation of nonlinear systems and allows to model the response to multisine inputs. Finally, the Higher Order Sinusoidal Describing Functions (HOSDF) model the response to sinusoidal signals at both the fundamental excitation frequency and higher harmonics of this frequency. These describing functions are defined for different classes of input signals and nonlinear systems and capture a different subset of nonlinear effects. In this section three describing functions are introduced in detail: the SIDF, GDF and the Higher Order Sinusoidal Input Describing Function (HOSIDF).

First, consider the SIDF, which is defined for sinusoidal input signals and is defined as follows:

Definition 6 ($\mathfrak{D}_S(\xi, \gamma)$): Sinusoidal input describing function (SIDF)). Consider a nonlinear system² (11), with $y(t) \in \mathbb{R}$, subject to a sinusoidal input $u \in \mathbb{S}$ with frequency ξ_0 and amplitude γ and

² The authors in [17] define these describing functions for systems that 'have a single loop configuration with separable linear and nonlinear elements', here de-

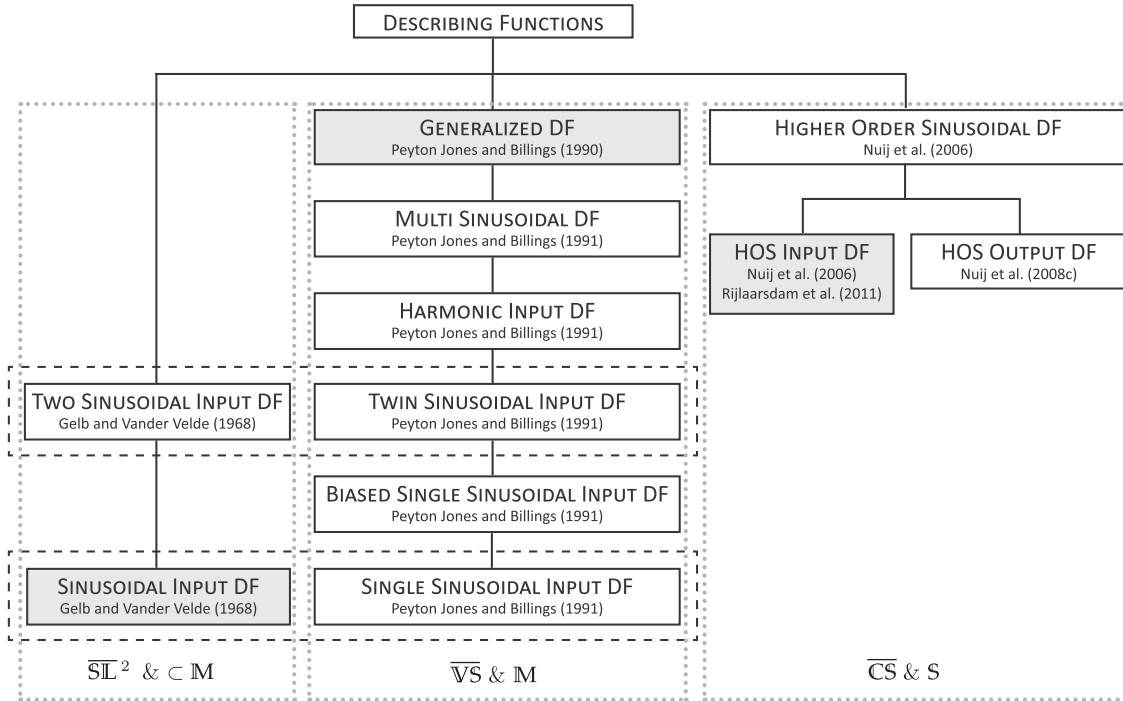


Fig. 3. Overview of a selection of different describing functions. The signal and system classes refer to the definitions used by the authors in the corresponding references.

define the systems steady state output $y(t)$ and Fourier transforms of the input and output $\mathcal{U}_F(\xi)$, $\mathcal{Y}_F(\xi) \in \mathbb{C}$. Then the sinusoidal input describing function $\mathcal{D}_S(\xi, \gamma) : \mathbb{R}_{>0} \times \mathbb{R} \mapsto \mathbb{C}$ is defined as:

$$\mathcal{D}_S(\xi_0, \gamma) = \frac{\mathcal{Y}_F(\xi_0, \gamma)}{\mathcal{U}_F(\xi_0, \gamma)}$$

(see: [17])

The SIDF models the response to a sinusoidal input by relating the spectral components in the input and output at the fundamental excitation frequency. Hence, the SIDF captures gain compression / expansion (see Table 1), but fails to model desensitization as the input is a single sinusoid. Moreover, intermodulation is not captured for the same reason and generation of harmonic components is not detected as only the response at the excitation frequency is considered.

The GDF, on the other hand, captures a wider range of nonlinear phenomena and allows for a broader class of input signals, but is defined only for Volterra systems. Consider the following definition of the GDF:

Definition 7 ($\mathcal{D}_G(\xi, \varsigma)$): Generalized descr. function (GDF). Consider a Volterra system (11) subject to multisine input $u \in \mathbb{M}$ scaled by a factor ς and output $y(t)$. Moreover, define the corresponding Fourier transforms of the input and output $\mathcal{U}_F(\xi)$, $\mathcal{Y}_F(\xi) \in \mathbb{C}$. Then, the generalized describing function $\mathcal{D}_G(\xi, \varsigma) : \mathbb{R}^2 \mapsto \mathbb{C}$ is defined as:

$$\mathcal{D}_G(\xi', \varsigma) = \frac{\sum_{p=1}^{\infty} \varsigma^{p-1} \mathcal{Y}_{F_p}(\xi')}{\mathcal{U}_F(\xi')}$$

where the GDF is defined for all $\xi' \in \{\xi \in \mathbb{R} \mid |\mathcal{U}_F(\xi)| \neq 0\}$ and the uni-dimensional spectrum $\mathcal{Y}_{F_p}(\xi')$ is related to the output gener-

ated by the order p th order GFRF, $\mathfrak{T}_p(\omega_p)$ by:

$$\mathcal{Y}_{F_p}(\xi') = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \mathfrak{T}_p \left(\xi_1, \dots, \xi_{p-1}, \xi' - \sum_{m=1}^{p-1} \xi_m \right) \cdot \mathcal{U}_F \left(\xi' - \sum_{m=1}^{p-1} \xi_m \right) \prod_{\ell=1}^{p-1} \mathcal{U}_F(\xi_\ell) d\xi_1 \dots d\xi_{p-1}$$

(see: [60])

The GDF allows to model the response of Volterra systems to multisine input signals [59,60]. However, it is specific to the selected input signal, which limits the validity of the model to the excitation signal used. As the GDF allows for broadband input signals, both gain compression / expansion and desensitization are captured by the GDF. However, intermodulation and the generation of harmonics are not modeled as the GDF only models the response at frequencies present in the input signal.

Finally, the Higher Order Sinusoidal Input Describing Functions model the response of uniformly convergent nonlinear systems to a sinusoidal input at harmonics of the excitation frequency and are defined as follows:

Definition 8 ($\mathfrak{H}_k(\xi, \gamma)$): Higher order sinusoidal input describing function (HOSIDF). Consider a uniformly convergent, time invariant nonlinear system (11) subject to a sinusoidal input $u \in \mathbb{S}$ with frequency ξ_0 and amplitude γ . Next, define the systems steady state output $y(t)$ and single sided spectra of the input and output $\mathcal{U}(\xi)$, $\mathcal{Y}(\xi) \in \mathbb{C}$. Then, the k th higher order sinusoidal input describing function $\mathfrak{H}_k(\xi, \gamma) : \mathbb{R}_{>0} \times \mathbb{R} \mapsto \mathbb{C}$, $k = 0, 1, 2, \dots$ is defined as:

$$\mathfrak{H}_k(\xi_0, \gamma) = \frac{\mathcal{Y}(k\xi_0, \gamma)}{\mathcal{U}^k(\xi_0, \gamma)} \quad (8)$$

where $\mathcal{U}^k(\xi_0, \gamma) = \prod_{\ell=1}^k \mathcal{U}(\xi_0, \gamma)$.

(see: [46,71])

Remark. Note that $\mathfrak{H}_1(\xi_0, \gamma) = \mathcal{D}_S(\xi_0, \gamma) \forall \xi_0 > 0$.

noted by $\mathbb{SL}$. However, this describing function can be computed for all system classes considered in this paper.

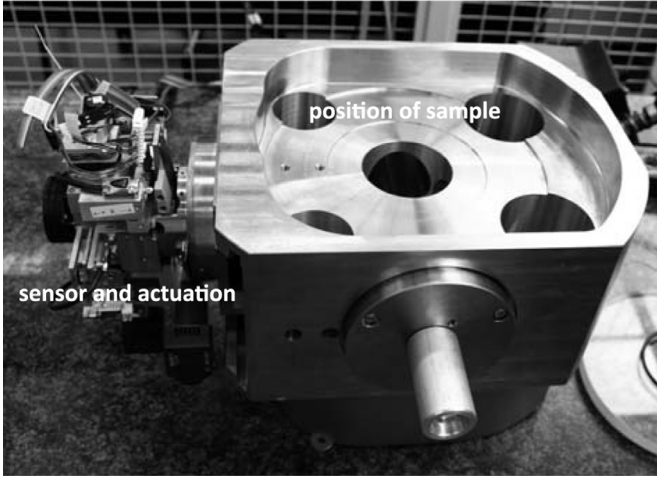


Fig. 4. Industrial high precision stage used in a transmission electron microscope.

Remark. Compared to the original definition in [46], the amplitude dependency $|\hat{s}_k| \propto \gamma^k$ inherent to the original HOSIDF model structure is removed in [71] such that the HOSIDFs reveal only system characteristics.

The HOSIDFs model the response to a sinusoidal input signal at harmonics of the fundamental input frequency. Therefore both gain compression / expansion and the generation of harmonics are captured in this describing function. However, as the excitation signal is a single sinusoid, both desensitization and intermodulation are not captured by the HOSIDFs. The HOSIDFs yield valuable information about nonlinearities that can be used for modeling and control purposes [47,48,71,72]. Finally, related studies as presented in [49] use the concept of the HOSIDF to find an input signal that yields a sinusoidal output signal, similar to Volterra based approaches presented in [32,34].

A comparison of the describing functions discussed in this section is provided in Table 1. Next, this section is concluded by an example that illustrates the identification of the HOSIDFs in practice.

Example 4 (experimental identification of the HOSIDFs). Consider an industrial high precision stage (Fig. 4) used in a Transmission Electron Microscope (TEM). At the high resolution and reproducibility that is required in a TEM, nonlinear effects start to effect the performance of the system significantly. This motivated the measurements presented in this example. The motion stage is a single input, single output system with the voltage to the current amplifier as the input and the position of the stage as its output. The identification is performed in open loop as in [46].

The system is excited with a sinusoidal input and the response is measured using a SigLab 20 – 42 dynamic signal analyzer. To measure the response of the system an encoder system is used that allows high resolution position measurements over the entire stroke of the set-up. To measure a single HOSIDF $\hat{s}_k(\xi_0, \gamma)$ for a given frequency / amplitude combination, several sinusoidal input signals $u \in \mathbb{S}$ are applied. Repeating the experiment multiple times yields the average HOSIDF and variance on the average which are denoted by $\hat{s}_k(\xi_0, \gamma)$ and $\zeta^2(\xi_0, \gamma)$ respectively.

By repeating each experiment ten times with frequencies ranging from 5 [Hz] to 300 [Hz] in steps of 5 [Hz] and input signal powers $\gamma/\sqrt{2}$, ranging from 0.07 [V] to 1.41 [V] (logarithmically spaced) the HOSIDFs are identified. All measurements have been performed with a sampling frequency of 5120 [Hz] and a block length of 2048 points, resulting in leakage free measurements.

A typical series of HOSIDFs is depicted in Fig. 5a. Given the odd nature of the friction nonlinearity in this case, the even HOSIDFs are low. Fig. 5a hints that the system behaves more linearly for increasing value of γ . This can be observed from the combination of a decreasing gradient $\partial \hat{s}_1 / \partial \gamma$ and a decreasing value of $\hat{s}_k(\xi, \gamma) \forall k > 1$ for all measured frequencies. Finally, in Fig. 5b $|\hat{s}_1|$ is depicted as a function of input power and frequency.

6. Linear approximation in the presence of nonlinearities

In many applications, the use of LTI models is common practice, mainly because methods to identify and interpret such models are well developed [44] and many tools exist that allow controller design and synthesis based on LTI models [86]. However, most real life systems are not linear and it is therefore required to assess the validity of a linear approximation (see for example [14,63,79,82]). In this section, the estimation of a linear model in the presence of nonlinearities is addressed.

Consider a nonlinear system as depicted in Fig. 6a and assume the system can be transformed to a structure as in Fig. 6b. In Fig. 6b, the output $\mathcal{Y}_F(\xi)$ is assumed to exist of a component that is generated by an LTI system $\mathfrak{B}_u(\xi)$, a distortion $\mathcal{S}_{F_u}(\xi)$ due to nonlinearities and an output disturbance $\mathcal{N}_F(\xi)$. Note that the transformation in Fig. 6 is nontrivial and a (unique) transformation may not exist. In the sequel, the existence and invariance of this transformation is addressed. However, for now, consider the representation in Fig. 6b and note that the nonlinear system is represented by the combination of an LTI model $\mathfrak{B}_u(\xi)$ and a nonlinear disturbance $\mathcal{S}_{F_u}(\xi)$, both of which depend on the power spectrum of the input signal.

$\mathfrak{B}_u(\xi)$ is referred to as the Best Linear Approximation (BLA) [14,82]. A BLA can be computed for a large class of input signals and nonlinear systems, but additional assumptions are required to assure invariance of the approximation with respect to a given class of input signals. In this section the BLA is therefore introduced in two steps. First, the formal definition of the BLA is provided and then sufficient conditions are provided for a class of nonlinear systems and input signals for which the BLA exists and is invariant.

The transformation in Fig. 6 is defined by the pair $(\mathfrak{B}_u, \mathcal{S}_{F_u})$, which is defined along the lines of [15,82].

Definition 9 ($(\mathfrak{B}_u, \mathcal{S}_{F_u})$: Best Linear Approximation). Consider a nonlinear system (11), with $y(t) \in \mathbb{R}$ and assume the transformation in Fig. 6 exists, such that:

$$\mathcal{Y}_F(\xi) = \mathfrak{B}_u(\xi)\mathcal{U}_F(\xi) + \mathcal{S}_{F_u}(\xi) + \mathcal{N}_F(\xi)$$

then, the Best Linear Approximation (BLA) is defined by the pair $(\mathfrak{B}_u, \mathcal{S}_{F_u})$, such that:

$$\mathfrak{B}_u(\xi) = \arg \min_{\mathfrak{B}(\xi)} E\{ (y(t) - \mathfrak{B}(\xi)u(t))^2 \} \quad (9)$$

where the expected value $E\{\cdot\}$ is the ensemble average over the considered class of inputs.

(see: [15])

Definition 9 does not assure the existence of the BLA, i.e. of the minimizer in (9). Moreover, although it is clear that the BLA depends on the power spectrum of the input signal, invariance of the BLA with respect to a class of input signals needs to be addressed as well. Therefore, consider the following class of excitation signals.

Definition 10 ($\mathbb{G}_{eq}$: Gaussian Riemann equivalent signals). Consider a power density spectrum $S_u(\xi)$, which is piecewise continuous and has a finite number of discontinuities and consider an excitation signal $u(t)$ that equals:

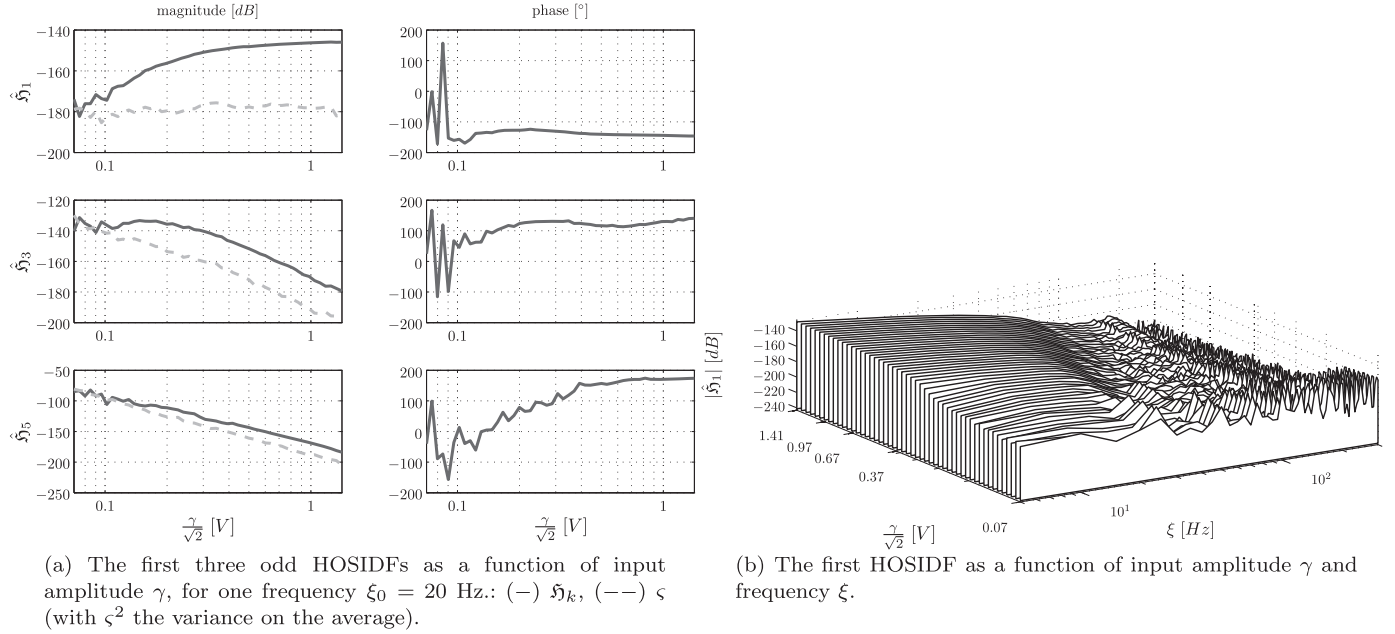


Fig. 5. Measured HOSIDF of the motion stage in a TEM.

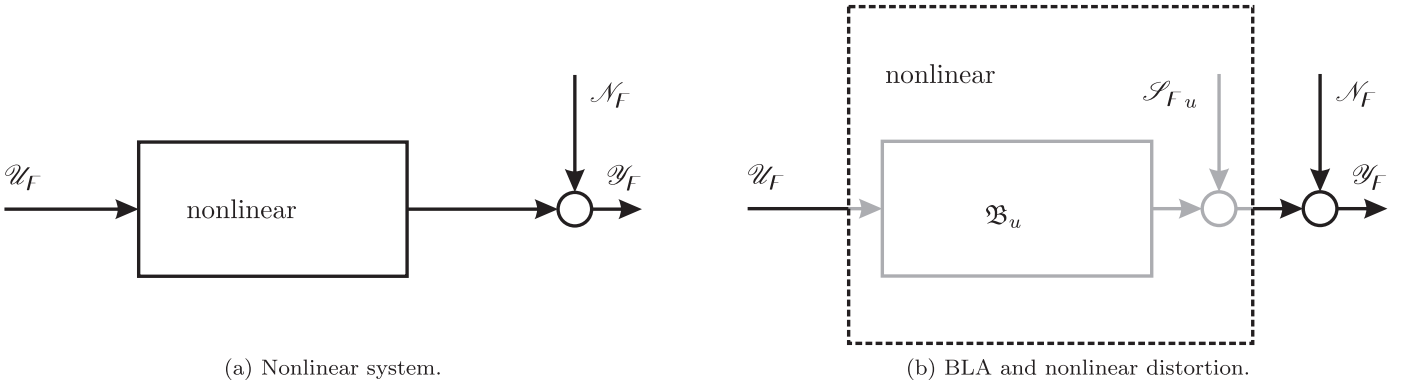


Fig. 6. Equivalent representations of a nonlinear system.

1. A Gaussian noise excitation $\mathbb{G}$ with power density spectrum $S_u(\xi)$, or
2. A periodic, real valued signal $u(t) \in \mathbb{R}$ such that $\mathcal{U}_F(\pm\xi_k) = A_k e^{\pm i\varphi_k}$, $k \in \mathbb{N}_{>1}$, $\xi_k \in \mathbb{R}_{>0}$ and $\mathcal{U}_F(\xi) = 0$ otherwise. Moreover, A_k and φ_k are random and mutually independent, $E\{e^{i\varphi_k}\} = 0$ and either $E\{A_k^2\} = f_A^2(\xi_k)$ or $A_k^2 = f_A^2(\xi_k)$ with $f_A(\xi)$ a piecewise continuous function with a finite number of discontinuities.

Then, the sets of signals 1 and 2 are Gaussian Riemann equivalent if:

$$\sum_{k \in \mathbb{K}} E\{|\mathcal{U}_F(\xi_k)|^2\} = \int_{\zeta_1}^{\zeta_2} S_u(\xi) d\xi + O(N_{\mathbb{K}}^{-1}) \quad \forall \zeta_i \in \mathbb{R}_{>0}$$

with $\mathbb{K} = \{k \in \mathbb{N}_{>1} | \zeta_1 < \xi_k < \zeta_2\}$, $N_{\mathbb{K}}$ the number of elements in $\mathbb{K}$, $0 < \zeta_1 < \zeta_2$ and where $E\{\cdot\}$ denotes the expected value.

(proof: [82])

To assure that the minimizer in (9) and thus the BLA exists, the class of nonlinear systems is restricted as follows (along the lines of [14]):

Definition 11 ($\overline{\text{WS}}$: special Wiener system). System (11) is called a special Wiener system if there exists a Volterra series representation (2), with continuous multidimensional Fourier transforms (3), that converges in mean square sense, with probability 1 to $y(t)$ for

all bounded inputs $u \in \mathbb{G}_{eq}$, i.e.

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau E\{|y(t) - y_p(t)|^2\} dt = 0 \quad \forall \tau \in \mathbb{R}_{\geq 0}$$

where the expected value $E\{\cdot\}$ is the ensemble average over the considered class of random inputs.

Remark. This class of systems is referred to as special Wiener systems as it is closely related to the classical Wiener systems [78] for which the requirement on continuity of the Fourier transforms is not imposed and which considers only Gaussian inputs $\mathbb{G} \subset \mathbb{G}_{eq}$ in the convergence criterion.

Using Definitions 10 and 11, the following sufficient conditions for the existence and invariance of the BLA are derived in [14].

Lemma 1 (Invariance and existence of the BLA). Consider a nonlinear system (11). If the system:

1. Belongs to the class of special Wiener systems $\overline{\text{WS}}$;
2. Is subject to Gaussian Riemann equivalent input signals, i.e. $u \in \mathbb{G}_{eq}$,

then, the best linear approximation (Definition 9) exists and the pair $(\mathfrak{B}_u, \mathcal{S}_{Fu})$ is asymptotically invariant under inputs in the equivalence class $\mathbb{G}_{eq}$. Moreover, if condition 1–2 are satisfied, the BLA

equals:

$$\mathfrak{B}_u(\xi) = \frac{E\{\mathcal{Y}_F(\xi)\mathcal{U}_F^*(\xi)\}}{E\{\mathcal{U}_F(\xi)\mathcal{U}_F^*(\xi)\}} \quad (10)$$

where $*$ indicates the complex conjugate and the expected value $E\{\cdot\}$ is the ensemble average over the considered class of inputs which includes averaging over the effect of the measurement noise $\mathcal{N}_F$.

Proof. (see³ [14]) $\square$

Remark. Although Lemma 1 assures invariance of the BLA, the BLA still specific to the fixed power spectrum of the set of input signals that is used.

Remark. In the sequel, the BLA is considered under the conditions in Lemma 1 which assures both invariance and existence of the linear approximation but does not provide any condition about the quality of the approximation.

Experimental identification of the BLA and the detection of nonlinear effects in the frequency domain requires careful design of the excitation signal and signal processing. An overview of techniques for the detection of nonlinearities is presented in [91]. Moreover, excitation signal design is for example addressed in [19,20,79] and signal processing techniques are discussed in for example [63,66,67,84].

The concept behind the detection of nonlinear effects in the frequency domain is often based on two approaches:

1. The presence of nonlinearities will cause a change in linear approximation of the dynamics when the input signal changes. Hence, a variance analysis on an ensemble of appropriately selected input signals allows to quantify the effects of nonlinearities [13,90].
2. Nonlinearities can be detected by analyzing the power appearing in the output spectrum at spectral lines that are not excited by the input signal [62,68].

Combining these approaches allows quantification of nonlinear effects and characterization of the nature of the nonlinearities. A detailed discussion of these methods is outside the scope of this work but an example is presented at the end of this section that illustrates both methods.

Both the identification of the BLA and means to detect nonlinearities are often discussed for single input, single output systems that are identified in open loop. However, extensions for multi input, multi output systems are presented in [81,83]. Moreover, identification of the BLA and estimation of nonlinear effects in closed loop is addressed in for example [61,65,69]. Finally, the following example concludes this section and illustrates the identification of the BLA in practice.

Example 5 (Experimental identification of a high precision motion stage). Consider again the high precision motion stage from a transmission electron microscope as in Example 4. In the following both the BLA of this system is identified and the effects of nonlinearities on the dynamics are quantified.

First consider a conventional identification experiment using band limited white noise with three different root mean square (rms) values. The estimates depicted in Fig. 7 clearly indicate nonlinear behavior as the FRF changes with the rms value of the input signal. This indicates that further analysis of the validity of a linear model is required. Therefore, the BLA is identified in a setting that allows to quantify nonlinear effects and assess the validity of this linear approximation.

In this example the BLA is identified under the assumptions in Lemma 1. Hence, the system is assumed to be in the class of

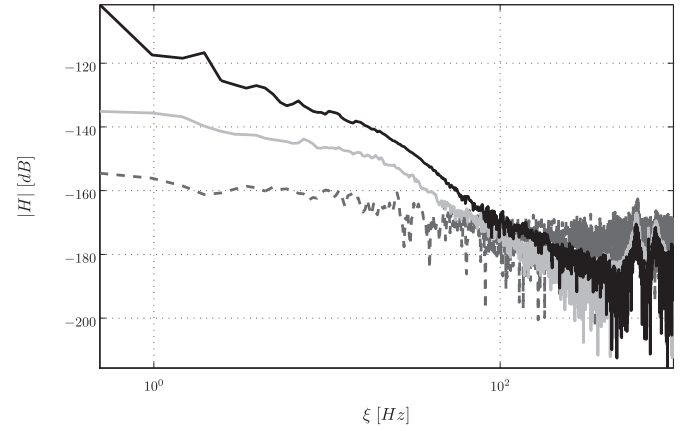


Fig. 7. Linear approximation identified with different powers of the input signal (–) $u_{rms} = 0.10$ [V], (– grey) $u_{rms} = 0.50$ [V] and (– black) $u_{rms} = 1.0$ [V].

Table 2
Estimated variables (see [70]).

Variable	Description
$\mathfrak{B}_u$	Best linear approximation
$\mathcal{Y}_F$	Output spectrum
$\sigma_{\mathcal{N}}^2$	Variance on $\mathcal{Y}_F$ due to noise
$\sigma_{\mathcal{F}}^2$	Variance on $\mathcal{Y}_F$ due to nonlinearities
$\mathcal{P}(\ell_{o/e})$	Power at non-excited odd (o) and even (e) lines

$\overline{\text{WS}}$ systems. Then, application of Gaussian Riemann equivalent input signals, yields an invariant estimate of the BLA for a given power spectrum of the input signal. In the following, random odd multisine excitation signals are used [70], which are Gaussian Riemann equivalent.

The identification procedure consists of exciting the system with several periods of different realizations of a random multisine. Computation of the averages and variances then yields the BLA and an estimate of nonlinear influences based on: 1. Variance information and 2. Energy appearing at non excited lines. The signal processing involved is discussed in [70] and yields the estimated variables in Table 2.

Ten realizations of the odd random phase multisine have been generated and the response has been measured for ten consecutive periods. Furthermore, this experiment is repeated for twenty different rms values of the multisines which are logarithmically scaled between 0.3 [V] and 5.0 [V]. Measurements are performed using a SigLab measurement system with a measurement frequency of 2560 [Hz] and a block length of 8192 measurement points. The multisine signal is comprised of sinusoid with frequencies $k\xi_0 = 0.3125$ [Hz], $k \in \mathbb{N}$. Finally, signal processing as in [70] is applied to each measurement set, yielding a typical output spectrum as depicted in Fig. 8a.

Fig. 8a shows that the power generated by nonlinear effects has an average level that is 10 [dB] lower than the power generated in the output spectrum by the BLA of the system. Analysis of the non-excited lines shows that the nonlinearities have both an odd and even nature, but the odd components dominate by almost 20 [dB]. Furthermore, it becomes clear that the variation due to nonlinearities is of the same order of magnitude as the odd nonlinearities that are detected. Finally the variation due to process and / or measurement noise is almost 30 [dB] lower than the variation due to nonlinear effects.

Fig. 8b depicts the BLA of the system as a function of both the rms value and frequency of the input. This figure indicates that for high input power, the system behaves more linear. This becomes clear from the reducing gradient $\partial|\mathfrak{B}_u|/\partial u_{rms}$.

³ See also [63,64,80].

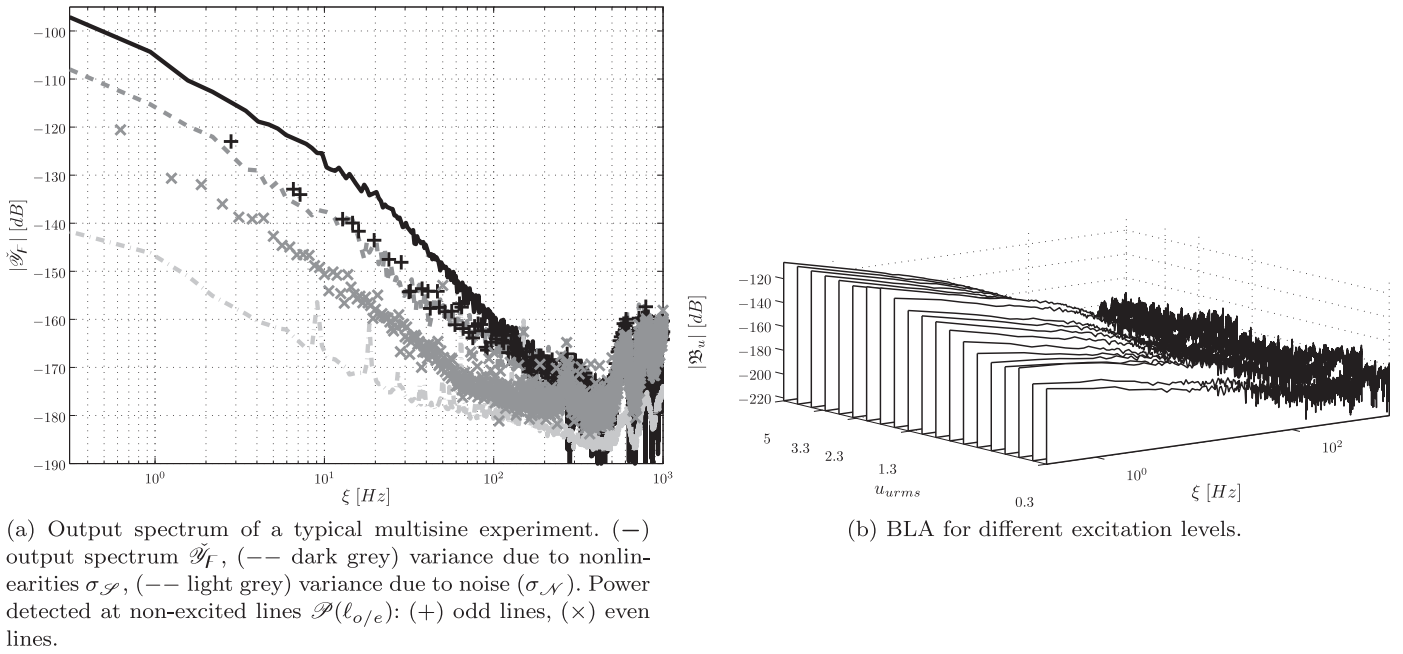


Fig. 8. Estimated output spectrum and BLA for different excitation levels.

7. Comparison

In this section a comparison between the different frequency domain models and the corresponding signal and system classes is provided. First, the different system and signal classes for which the models are defined are compared. Next, a qualitative comparison between the frequency domain models is provided yielding an overview of the applicability of the different models and the nonlinear effects that can be detected using each of the model types.

System and signal classes

In Fig. 9 an overview of the signal and system classes discussed in this paper is provided. Both random and deterministic signals are considered. The relation between different signal classes is as follows: By definition, the class of sinusoidal signals (Definition 16) is a subset of the class of multisine signals (Definition 15). Similarly, the class of Gaussian signals (Definition 14) is a subset of the class of Gaussian Riemann equivalent signals (Definition 10).

Next, consider the comparison of the different system classes depicted in Fig. 9b. The classes of Volterra and special Wiener systems can directly be compared as both prescribe the same type of mathematical model. The classes of Volterra and special Wiener systems partially overlap, but neither is a subset of the other, e.g. Volterra systems with discontinuous Fourier transforms are in the class of special Wiener systems. The class of Wiener systems includes the classes of Volterra and special Wiener systems as the continuity condition on the Fourier transform is relaxed in this case.

The class of convergent systems can not be straightforwardly be related to the classes of Volterra and special Wiener systems. Convergence is a property of a state space realization of a system and concerns the existence and stability of a unique limit solution of the nonlinear system. This requires the definition of initial conditions and a well defined notion of a limit solution. As the Volterra series does not include the definition of an initial condition it can not straightforwardly be related to the notion of convergence.

On a side note, in [11] a concept similar to convergence, called Fading Memory (FM), is introduced. On the one hand, convergence relates to the existence, stability and uniqueness of a steady state solution, which is independent of initial conditions/ Hence, convergence is concerned with the input to state properties of a state space realization. FM, on the other hand, relates to the dependence of the steady state solution on the input signals themselves and is concerned with the input to output properties of an operator. For an operator that has a realization of some sort, the notion of FM may allow to proof the existence of a unique steady state solution for a subset of input signals and initial conditions [11]. However, the following two arguments illustrate why FM and convergence are not equivalent. First of all, an operator may not have a state space realization (11). Hence, even if an operator has FM, a state space realization may not exist, such that convergence cannot be claimed. Secondly, an operator which has a realization (11) that includes unstable non-observable dynamics may still have FM. However, the corresponding realization cannot be convergent.

Frequency domain models

This section provides a comparison of the frequency domain methods for nonlinear systems considered. First, consider the overview presented in Table 3. This table depicts the applicability of each model type for each combination of input signal and class of nonlinear systems. For all model types, apart from the SIDF, the definitions used in literature are used to generate Table 3.

The GFRF is applicable for all signal types considered, but is limited to modeling Volterra systems. The corresponding GDF is applicable to Volterra systems as well, but valid for multisine inputs only. A detailed analysis connecting the GFRF and the HOSIDFs can be found in [73].

For convergent systems the HOSIDF and NFRF allow to model the systems dynamics, but these are restricted to sinusoidal inputs. The SIDF can be computed for all system classes considered in this paper, but is restricted to sinusoidal inputs as well. Finally, the BLA when considered under the conditions in Lemma 1, is defined for special Wiener systems subject to Gaussian Riemann equivalent signals.

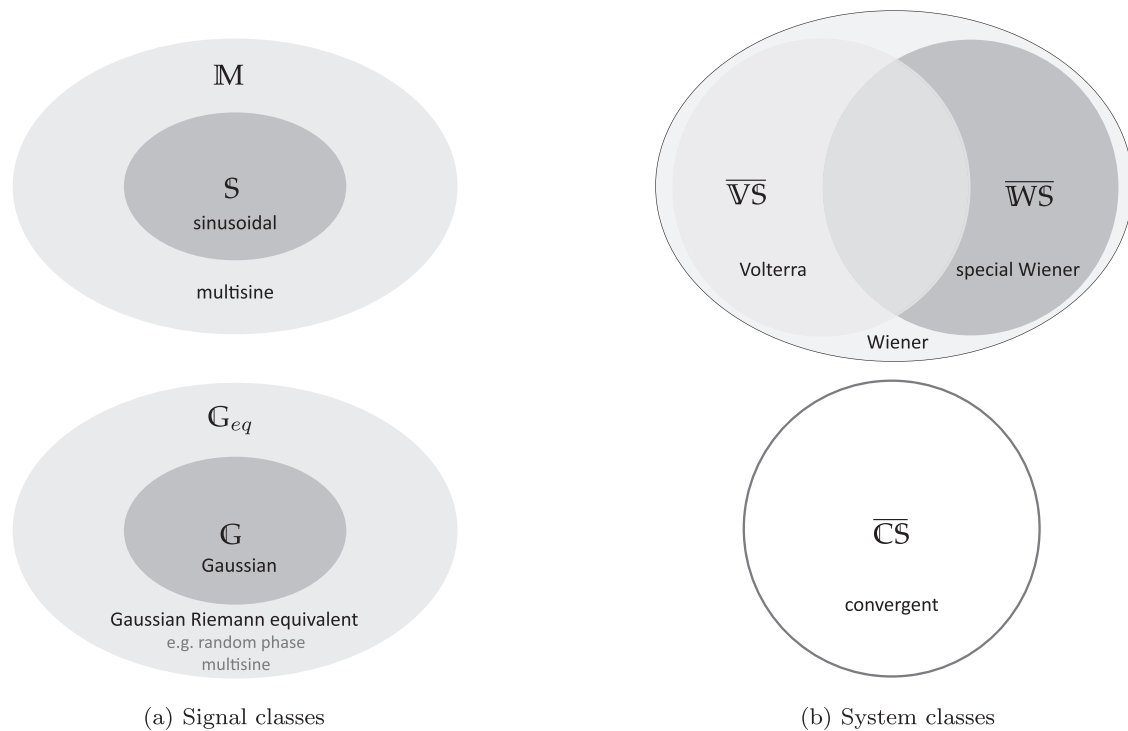


Fig. 9. Comparison of the system and signal classes.

Table 3

Model types ordered by signal and system class (according to definitions used in references).

Signal	Deterministic		Random
system	S (sinusoidal)	M (multisine)	G _{eq} (Gaussian Riemann eq.)
VS (Volterra)	GFRF GDF SIDF	GFRF GDF	GFRF
WS (Special Wiener)	SIDF		BLA
CS (Convergent)	HOSIDF NFRF SIDF		

Moreover, Table 1 provides an overview of the information present in each model type. To fully reconstruct the response of a nonlinear system to a given input signal the corresponding model needs to model both the response at excited and non excited frequencies and both phase and gain information needs to be recoverable from the model. This is the case for the GFRF and this model therefore allows to fully reconstruct the response of a given Volterra system to a class of input signals. The HOSIDF and NFRF also allow to fully reconstruct the steady state response of a uniformly convergent system to any given sinusoidal input signal although the representation of the nonlinear dynamics in both model types is different. However, the nonlinear Bode plot corresponding to the NFRF does not allow to reconstruct the full response as all phase information is lost in this representation. Finally, the SIDF, GDF and BLA do not allow to fully reconstruct the output of a nonlinear system as information about spectral components at non-excited frequencies is not available in these models.

Finally, Table 1 also presents an overview of the ability of the different models to detect the different effects of nonlinearities in the frequency domain. As becomes clear, all methods considered are able to model gain compression and expansion. Detection of desensitization however, requires multi-tone inputs and detection at excited spectral lines. Hence, only the GFRF, GDF and BLA are able to model desensitization. To model intermodulation, multi-tone in-

puts are also required, but detection must appear at non-excited spectral lines as well. Hence, the GFRF is able to capture intermodulation and the signal processing related to the BLA can be adapted to detect intermodulation. For the detection of higher harmonics a sinusoidal excitation suffices, but detection must appear at non-excited harmonic spectral lines as captured in the HOSIDF. Moreover, the GFRF models the generation of harmonics and these can be detected by adapting the signal processing related to the BLA as well. Finally, the appearance of harmonics in the systems output are captured by the NFRF, but the effects of gain compression / expansion and the generation of harmonics are hard to distinguish in the corresponding nonlinear Bode plot.

Applicability

In the preceding sections an overview and comparison of four main approaches to the frequency domain analysis of nonlinear systems was presented. To conclude this paper, a discussion concerning the application of the different methods is presented. The following reflects the personal opinion of the author and may serve as a guideline for those seeking to use any of the frequency domain methods discussed in this paper.

First of all, the GFRF provides a useful tool for the analysis of Volterra systems and allows to model the response to an extensive class of input signals. However, the practical application of the

GFRF is limited as identification of high order kernels / GFRFs is difficult. Moreover, even if a model is known, the GFRF may be used for prediction, but the model structure is complex, which limits its application to practical analysis or control design. Secondly, the nonlinear Bode plot provides means to assess the gain of a nonlinear system and may be used for sensitivity analysis as well [96]. However, this model is only valid for sinusoidal inputs and the corresponding NFRF can be hard to identify in practice.

Thirdly, the HOSIDF are easily identifiable and provide a fast method to assess nonlinear effects in practice. However, although the HOSIDF provide some means to assess performance degrading nonlinear effects, this model is valid for sinusoidal inputs only. This limits their (real-time) application when non sinusoidal inputs are required. Finally, the BLA provides means to attain a well defined linear approximation of the nonlinear dynamics as well as a measure of the quality of this linear approximation. This yields a useful method to identify LTI approximations of weakly nonlinear systems. However, if severe nonlinearities are present or small nonlinearities negatively influence the performance of the system, the BLA provides insufficient information to address or model these effects.

Summarizing, if an LTI approximation of the systems dynamics is required, identification of the BLA is recommended as it is easy to identify, yields a widely valid linear approximation and provides a corresponding quality measure. If more information about the nonlinear behavior is required, the nonlinear Bode plot allows for a gain and sensitivity analysis of the nonlinear system. Finally, if even more information about the nonlinearity is required, the HOSIDFs provide gain and phase information about the nonlinear behavior, allowing analysis and control of the nonlinear effects.

8. Conclusion

This paper addresses the comparison of four well established frequency domain methods for nonlinear systems: Volterra based models, nonlinear frequency response functions, describing functions and linear approximations in the presence of nonlinearities. The main contributions are summarized in Tables 1 and 3, which compare the information captured in each modal and the signal / system classes for which each model is valid. The consistent introduction of each model, accompanied by (experimental) examples and their comparison provides insight into the differences and similarities between alternative approaches and their applicability in practice.

Appendix A. Nomenclature and definitions

In the following, signals are real-valued, scalar and denoted by non-capitalized roman letters, e.g. $u(t) \in \mathbb{R}$. Frequency domain representations of time domain signals, such as the Fourier transform and other spectra are denoted by corresponding capitalized calligraphic font, e.g. $\mathcal{U}(\xi) \in \mathbb{C}$. Furthermore, transfer functions are denoted by Fraktur font, e.g. $\mathfrak{H}(\xi) \in \mathbb{C}$ (unless specified otherwise). Moreover, signal classes are denoted by single blackboard bold capital letter, e.g. $\mathbb{S}$ and system classes by multiple, overlined blackboard bold capital letter, e.g. $\overline{\mathbb{S}}$. The sets of real, natural and complex numbers are denoted as usual by $\mathbb{R}$, $\mathbb{N}$ and $\mathbb{C}$ respectively and finally, differentiation with respect to time is denoted by $\dot{\cdot}$, i.e. $\dot{x}(t) = \frac{d}{dt}x(t)$.

System class

Let $\mathbb{B}_p$ denote the space of piecewise continuous, bounded functions⁴ $\mathbb{R}_{\geq t_0} \mapsto \mathbb{R}$, $t_0 \in \mathbb{R}$. In the following all systems considered are

continuous time, causal, time invariant and are assumed to have a realization of the form:

$$\begin{aligned} \dot{x}(t) &= f(x(t), w(t)) & x(t_0) &= x_0 \\ y(t) &= g(x(t), w(t)) \end{aligned} \quad (11)$$

where $x(t) \in \mathbb{R}^n$ are the states, $y(t) \in \mathbb{R}^m$ the outputs, $w \in \mathbb{B}_p$ the input of the system and $f: \mathbb{R}^n \times \mathbb{R} \mapsto \mathbb{R}^n$ and $g: \mathbb{R}^n \times \mathbb{R} \mapsto \mathbb{R}^m$. A solution of (11), corresponding to an input $w(t)$, is denoted by $x_w(t)$. The corresponding limit solution (if it exists) is denoted by $\tilde{x}_w(t)$, such that $\lim_{t \rightarrow \infty} \|x_w(t) - \tilde{x}_w(t)\| = 0 \forall x_0 \in \mathbb{R}^n$. Furthermore, the output corresponding to the limit solution is denoted by $\tilde{y}_w(t) = g(\tilde{x}_w(t), w(t))$. Finally, a system (11) is called Linear and Time Invariant (LTI) if it satisfies the principles of superposition and homogeneity [101].

Signal classes

The analysis presented in the sequel makes use of several signal classes and transforms. Hence, consider a real valued signal $z(t) \in \mathbb{R}$ and define the corresponding Fourier transform as follows:

Definition 12 ($\mathcal{Z}_F(\xi)$: Fourier transform). Consider a signal $z(t)$. Then its Fourier transform $\mathcal{Z}_F(\xi)$ is given by:

$$\mathcal{Z}_F(\xi) = \int_{-\infty}^{\infty} z(t) e^{-2\pi i \xi t} dt \quad (12)$$

with $\xi \in \mathbb{R}$ the frequency in [Hz], $i = \sqrt{-1}$ and the corresponding scaling of the transform equal to one. The corresponding inverse Fourier transform is then given by:

$$z(t) = \int_{-\infty}^{\infty} \mathcal{Z}_F(\xi) e^{2\pi i \xi t} d\xi \quad (13)$$

Moreover, for real valued signals, all information available in the Fourier transform is captured in the single sided spectrum, i.e.

Definition 13 ($\mathcal{Z}(\xi)$: single sided spectrum). Consider a signal $z(t) \in \mathbb{R}$ and its Fourier transform $\mathcal{Z}_F(\xi)$. Then, the single sided spectrum $\mathcal{Z}(\xi)$, equals:

$$\mathcal{Z}(\xi) = \begin{cases} 2\mathcal{Z}_F(\xi) & \xi > 0 \\ \mathcal{Z}_F(\xi) & \xi = 0 \\ 0 & \xi < 0 \end{cases}$$

In the following, several signal classes are used. The classes of Gaussian, multisine and sinusoidal signals are of particular importance and are therefore defined below. First consider the sequence $z(t_k)$, which equals a Gaussian signal if it is defined as follows:

Definition 14 ($\mathbb{G}$: Gaussian signal). Let $\mathbb{G}$ denote the set of Gaussian signals. A continuous time signal $z(t)$ is said to be a Gaussian random process if, for every set of fixed times $\{t_n\}$, the random variables $x(t_n)$ follows a multidimensional normal distribution with a user defined power density spectrum $S_z(\xi)$.

Next, consider a real valued signal $z(t)$ which is composed of a finite number of sinusoidal components. In general such signal is referred to as a multisine and in the special case where the sinusoidal components are harmonically related, the resulting signal is called harmonic, i.e.

Definition 15 ($\mathbb{M}$: Multisine). Let $\mathbb{M}$ denote the set of multisine signals. A signal $z(t)$ is a multisine if $z(t) = \sum_{k=1}^K \gamma_k \cos(2\pi \xi_k t + \varphi_k)$, $t \in \mathbb{R}$, for some $\xi_k, \gamma_k \in \mathbb{R}_{>0}$, and $\varphi_k \in \mathbb{R}$. Furthermore, a signal $z(t)$ is called harmonic if it is a multisine and $\xi_k = k\xi_0$, $k \in \mathbb{N}$ for some $\xi_0 \in \mathbb{R}_{>0}$.

Finally, a special type of multisine signal is the single sine or sinusoidal signal, which is defined as follows:

Definition 16 ($\mathbb{S}$: Sinusoidal signal). Let $\mathbb{S} \subset \mathbb{M}$ denote the set of sinusoidal signals. A signal $z(t)$ is sinusoidal if $z(t) =$

⁴ Where $\mathbb{R}_{\geq t_0} = \{t \in \mathbb{R} | t \geq t_0\}$.

Table 4
Suggested references for each research area.

	GFRF	NFRF	DF	BLA
Background	[16] [18] [76] [78] [92] [93] [95]	[54]	[17]	[63]
Introduction	[98]	[55]	[17] [46] [59]	[63]
Analysis	[33] [38]	[52] [56]	[47] [60]	[81] [64] [80] [82] [14]
Applications	[6] [7] [8]	[96] [22] [25]	[71] [48] [49]	[13] [68] [75]

$\gamma \cos(2\pi\xi_0 t + \varphi_0)$, $t \in \mathbb{R}$, for some $\gamma, \xi_0 \in \mathbb{R}_{>0}$, $\varphi_0 \in \mathbb{R}$. Finally, $\mathbb{S}_{\xi_0} \subset \mathbb{S}$ denotes the subset of sinusoidal signals with frequency ξ_0 .

Appendix B. Literature overview

Table 4 provides an overview of the main literature related to each model.

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